

Week 9 - AI Regression Test Selection Agent Usecase - Architecture

Prompt 👍

AI Regression Test Selection Agent Architecture Design Prompt

Instruction:

Help me design an enterprise-grade architecture for an AI Regression Test Selection Agent.

Use the input and output requirements below to design the complete architecture.

Define:

- * High-Level Architecture
- * Low-Level Architecture
- * API Endpoints
- * Langflow Workflow
- * Data Flow
- * Component Interaction
- * Deployment Architecture
- * Security Considerations
- * Folder Structure
- * Future Enhancements

Input:

Release Notes

Jira Stories

Azure DevOps Work Items

Change Requests

User Stories

Bug Fixes

Technical Documentation

Confluence Pages

Wiki Pages

Regression Suite Metadata

Automation Scripts Repository

Historical Execution Reports

Defect Database

Production Incidents

Test Coverage Reports

Output:

1. Impacted Modules
2. Selected Regression Scripts
3. Excluded Regression Scripts
4. Risk Analysis Report
5. Coverage Gap Analysis
6. Recommended Additional Test Cases
7. Optimized Regression Execution Plan

8. JSON Output

9. CSV Output

10. Excel Output

11. REST API Output

Core Functional Requirements:

1. Release Change Understanding

Analyze:

- * Functional changes
- * Technical modifications
- * Impacted modules
- * Dependencies

2. Intelligent Regression Selection

Identify:

- * Relevant regression scripts
- * Unrelated scripts
- * Dependent modules

3. Risk-Based Prioritization

Prioritize based on:

- * Security impact
- * Business criticality
- * Customer-facing functionality
- * Historical failures

4. Coverage Gap Detection

Identify:

- * Missing automation coverage
- * Untested functionality
- * New feature coverage gaps

5. Regression Optimization

Reduce:

- * Execution effort
- * Execution time
- * Infrastructure cost

while maximizing:

- * Defect detection probability
- * Release confidence

Ingestion Pipeline:

Create embeddings for:

- * Release Notes
- * User Stories
- * Existing Regression Scripts
- * Historical Defect Data
- * Execution Reports

Retrieval Pipeline:

- * Hybrid Search
- * Semantic Search
- * Metadata Search
- * Reranking

- * Deduplication
- * Context Aggregation
- * Summarization

LLM Processing:

The LLM should:

- * Analyze release changes
- * Understand impacted functionality
- * Map release changes with regression scripts
- * Recommend relevant tests
- * Identify risk areas
- * Suggest new tests

Output Validation:

Validate:

- * Script relevance
- * Module mapping
- * Risk classification
- * Coverage recommendations

Vector Database:

MongoDB Atlas Vector Search

Embedding Model:

Mistral AI Embeddings

LLM Options:

- * GPT-4o
- * Claude
- * Gemini
- * Llama 3

Technology Stack:

Frontend:

- * React.js
- * Material UI

Backend:

- * Node.js
- * TypeScript
- * Express.js

Workflow Orchestration:

- * Langflow

Database:

- * MongoDB Atlas

Feedback Mechanism:

Support:

- * Approve
- * Reject
- * Edit

Store feedback for future optimization.

Folder Structure:

Design enterprise-level folder structure covering:

- * Frontend
- * Backend
- * Langflow
- * Vector Database
- * Prompt Management
- * Feedback System
- * Deployment

Context:

You are a Chief Enterprise Architect with expertise in:

- * Generative AI
- * Langflow
- * Software Testing
- * QA Automation
- * Microservices Architecture
- * Enterprise Platform Design

Persona:

Act as a Chief Architect responsible for designing an enterprise-grade AI-powered regression optimization platform.

Output Format:

Generate the architecture document in Markdown (.md) format.

The document must contain:

1. Executive Summary
2. Problem Statement
3. Business Challenges
4. Proposed Solution
5. High-Level Architecture
6. Low-Level Architecture
7. Langflow Workflow Design
8. Ingestion Pipeline
9. Retrieval Pipeline
10. LLM Processing Layer
11. API Design
12. Database Design
13. Security Architecture
14. Deployment Architecture
15. Folder Structure
16. Data Flow Diagram
17. Sequence Diagram
18. Technology Stack
19. Business Benefits
20. Future Enhancements

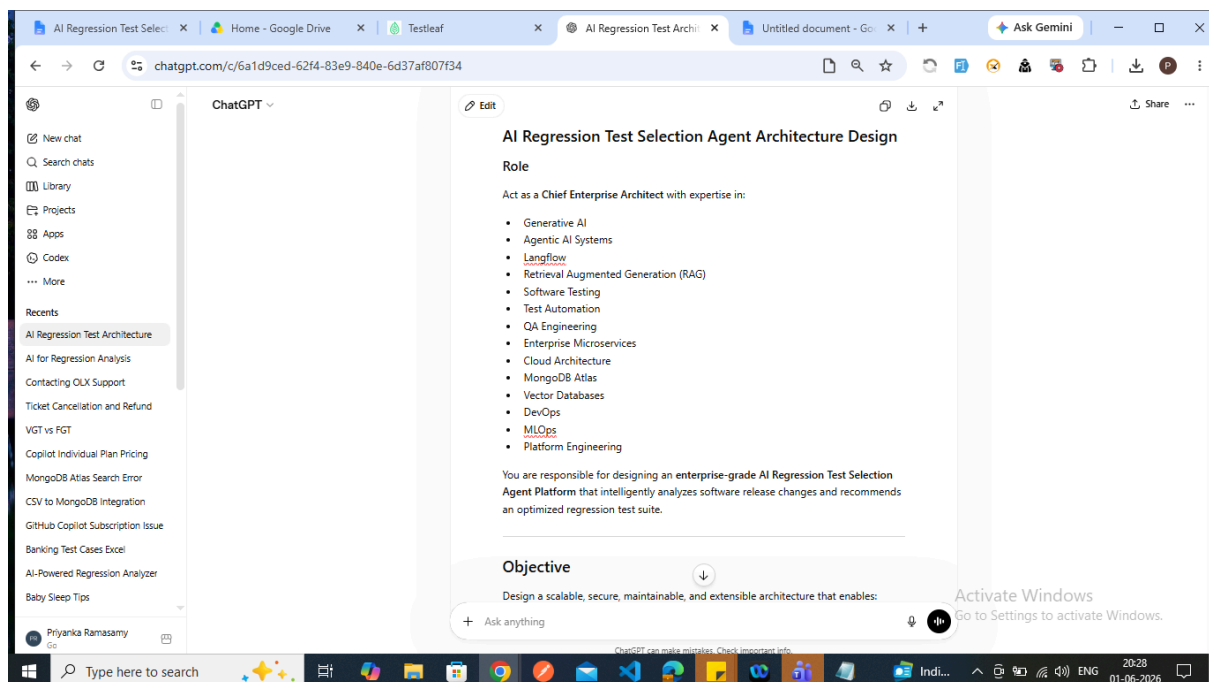
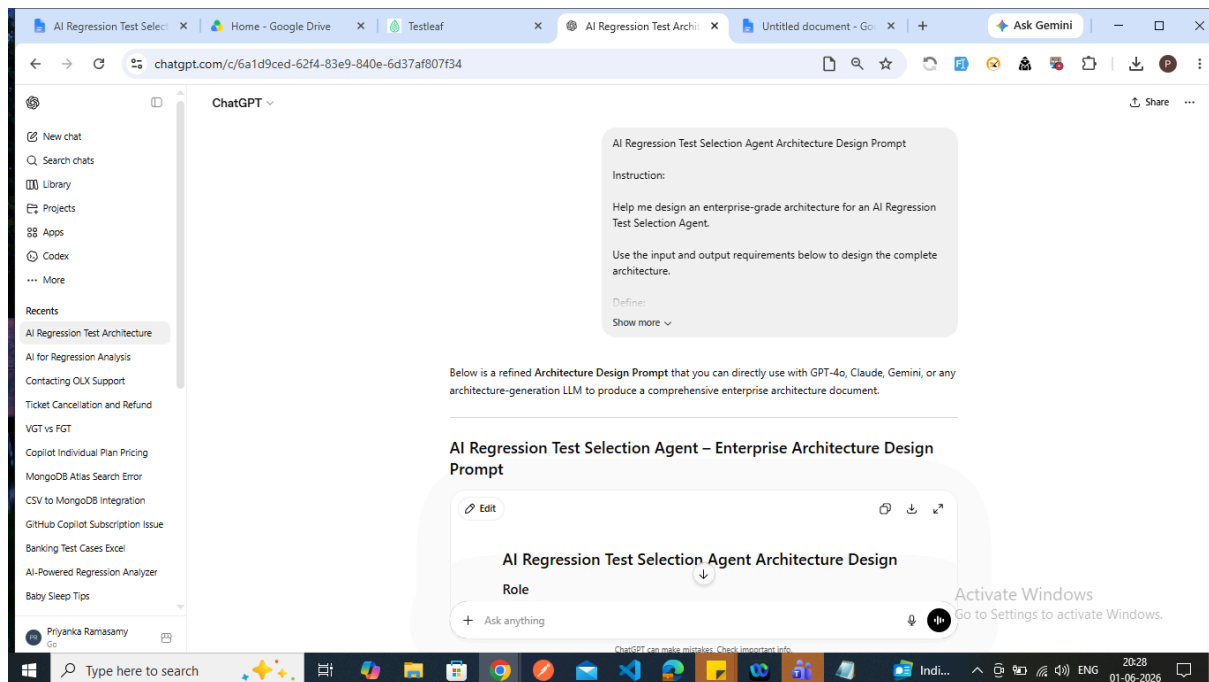
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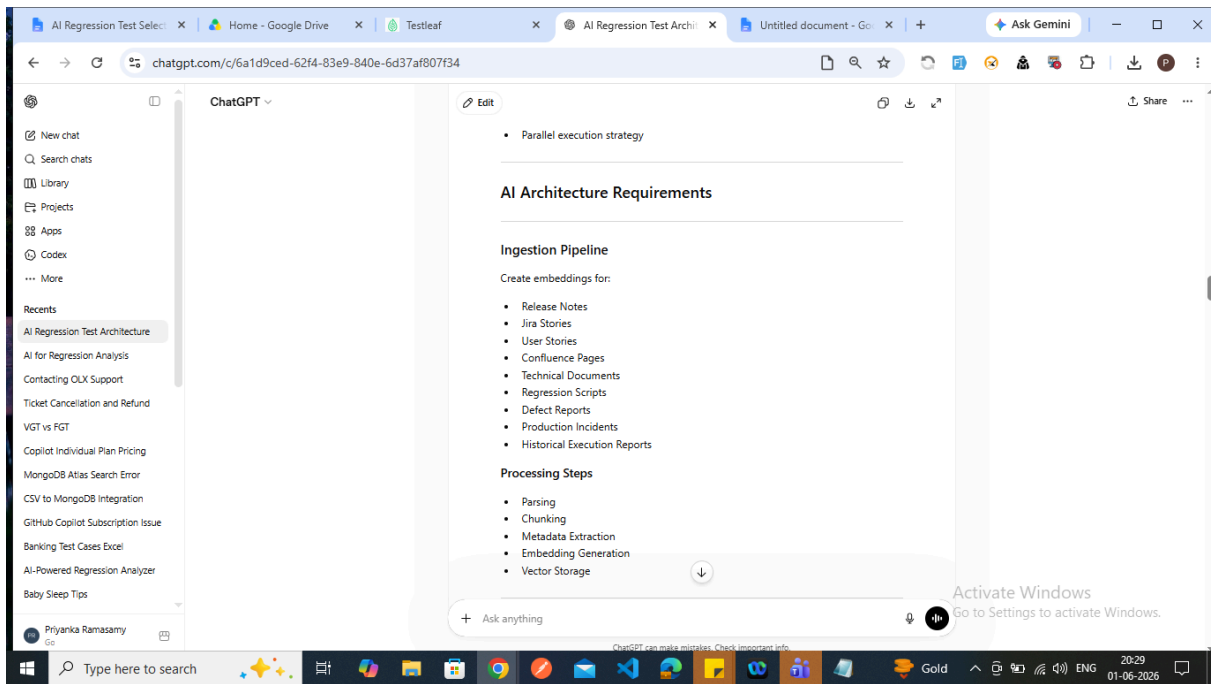
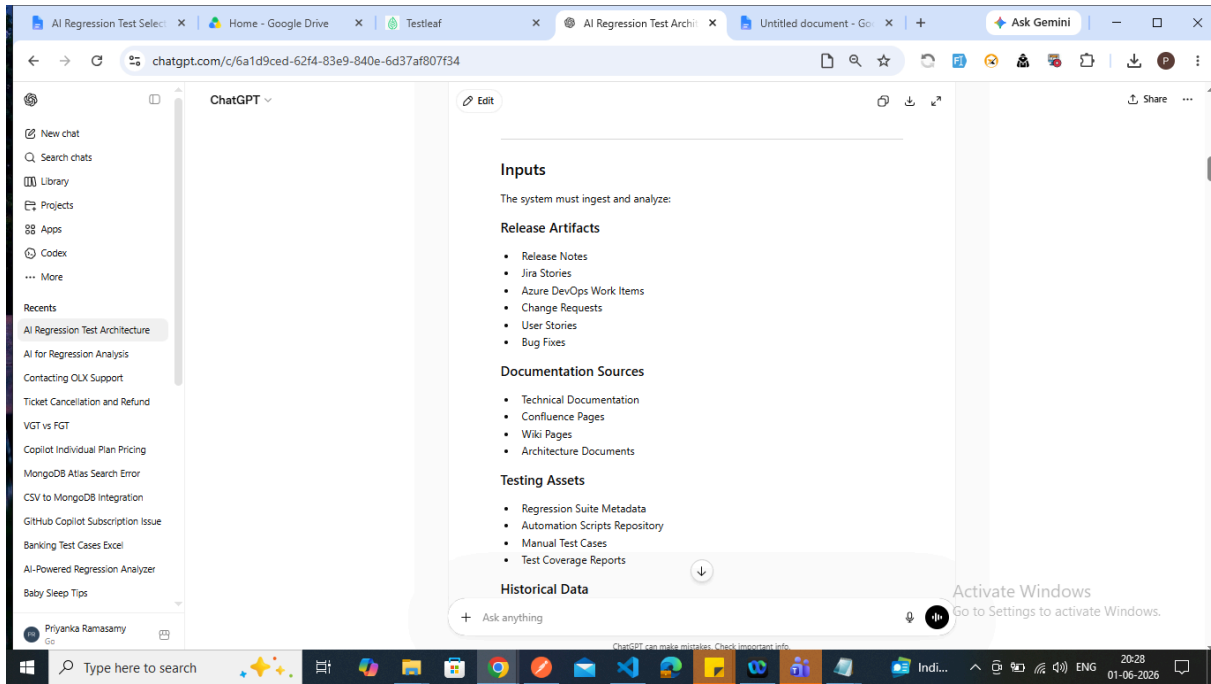
Formal, professional, enterprise architecture style.

Important Rules:

- * Do not hallucinate components.
- * Clearly justify every architectural decision.
- * Follow enterprise architecture best practices.
- * Include diagrams using ASCII format.
- * Include API request/response examples.
- * Include sample JSON output.
- * Design for scalability, maintainability, and extensibility.

Pasted the prompt in ChatGPT and generated the content





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Recents

- AI Regression Test Architecture
- AI for Regression Analysis
- Contacting OLX Support
- Ticket Cancellation and Refund
- VGT vs FGT
- Copilot Individual Plan Pricing
- MongoDB Atlas Search Error
- CSV to MongoDB Integration
- GitHub Copilot Subscription Issue
- Banking Test Cases Excel
- AI-Powered Regression Analyzer
- Baby Sleep Tips

Priyanka Ramasamy

Provide above content in downloadable .md format

I've created the Markdown file for download:

[Download AI_Regression_Test_Selection_Agent_Architecture_Prompt.md](#)

Note: The downloadable file contains a condensed version of the prompt structure. If you'd like the full expanded prompt exactly as provided above (all sections, requirements, APIs, security, deployment, Langflow workflows, etc.), I can generate a complete enterprise-grade .md file (several hundred lines) and provide that as a download as well.

Ask anything

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